

Behrooz Taheri

Power system protection

25 May 1993

Iran, Qazvin

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Google Scholar

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Skills

Team working

Digsilent

PSCAD

MATLAB

Latex

Office

Protection relay testing

Python

C#

C++

(*)[The skill scale is from 0 (Fundamental Awareness) to 6 (Expert).]

Research Interests

Power system protection, HVDC, IBR, and Artificial intelligence.

About me

I am Behrooz Taheri, born in Qazvin, Iran, in 1993. I earned both my M.Sc. (2018) and Ph.D. (2025) in Electrical Engineering from Qazvin Islamic Azad University (QIAU), where I specialized in power systems and renewable energy technologies. My Ph.D. research focused on advanced topics in electrical engineering, aiming to bridge cutting-edge theory with real-world application. Throughout my academic journey, I have been recognized twice as the top student researcher in Qazvin province, a testament to my dedication and passion for scientific excellence. I have authored and published over 40 scientific papers in renowned journals and conferences, making meaningful contributions to the field of power systems. My professional experience includes various industrial and research projects, particularly in applying Artificial Intelligence (AI) to enhance the efficiency, protection, and reliability of modern power systems. Since 2019, I have been an active member of the Young Researchers and Elites Club, fostering innovation and collaboration among emerging scientists. In 2024, I was honored to become a Senior Member of IEEE, reflecting my ongoing commitment to professional growth and contribution to the global engineering community. My core research interests include power system protection, transient analysis, microgrids, smart grids, Inverter-Based Resources (IBR), High Voltage Direct Current (HVDC), and the application of AI in renewable energy systemsparticularly in protection and control strategies to support a stable and sustainable energy future.

Education

2019-2025 Ph.D Electrical Engineering — Qazvin Islamic Azad University
Non-Intrusive Load Monitoring in Smart Home (Dr. Mostafa Sedighizadeh)

2015-2018 MS.c Electrical Engineering — Qazvin Islamic Azad University
Power Swing Detection Using rms Current Measurements (Dr. Farzad Razavi)

2012-2015 B.Sc Electrical Engineering — Darolfonoon University
Speed control of DC motor using PID controller using genetic algorithm (Dr. Hamideh Najafizadegan)

2010-2012 A.D Electrotechnics — Abhar Islamic Azad University
Research on a variety of solar power plants

Working Experience

2024-2025 Qazvin Islamic Azad University. — Qazvin, Iran
Instructor

2020-2024 Vebko Amirkabir research and development co. — Qazvin, Iran
Researcher

2020-2024 Buein Zahra Technical University (BZTE). — Qazvin, Iran
Instructor

2018-2019 Vebko Amirkabir research and development co. — Qazvin, Iran
Researcher

2015-2018 Vebko Amirkabir research and development co. — Qazvin, Iran
Electrical Engineering

Publications

Journal papers (English)

2025 B. Taheri, SA Hosseini, M Sedighizadeh, "Novel hybrid fuzzy-SVR model for fault detection in VSC-HVDC transmission lines,"International Journal of Electrical Power and Energy Systems.

2025 MM Ghalesefidi, B Taheri, M Yousefi, "A novel hybrid approach for power swing detection: a case study wind turbine,"Electrical Engineering.

2025 B. Taheri, SA Hosseini, "A Novel Hybrid Statistical Method for Power Swing Detection in Transmission Lines With GridFollowing Inverter,"IET Generation, Transmission & Distribution.

2025 B. Taheri, M. Sedighizadeh, M.R. Nasiri, A. Sheikhi Fini, "A non-intrusive load monitoring algorithm based on real-time feature extraction and deep learning model,"International Journal of Electrical Power and Energy Systems.

2025 B Taheri, SA Hosseini, H Hashemi-Dezaki, "Enhanced Fault Detection and Classification in AC Microgrids Through a Combination of Data Processing Techniques and Deep Neural Networks," Sustainability.

2024 SA Hosseini, B Taheri, SHH Sadeghi, A Nasiri, "A deep learning model for fault detection in distribution networks with high penetration of electric vehicle chargers," e-Prime-Advances in Electrical Engineering, Electronics and Energy.

2024 A. Sistani, SA. Hosseini, VS. Sadeghi, and B. Taheri, "Providing a fault detection method for the occurrence of faults in DC microgrids, distributed generations, and electrical vehicles," AUT Journal of Electrical Engineering.

2023 A. Sistani, SA. Hosseini, VS. Sadeghi, and B. Taheri, "Fault Detection in a Single-Bus DC Microgrid Connected to EV/PV Systems and Hybrid Energy Storage Using the DMD-IF Method," Sustainability.

2023 SA. Hosseini, B. Taheri, SHH. Sadeghi, and A. Nasiri, "An Overview of DC Microgrid Protection Schemes and the Factors Involved," Electric Power Components and Systems.

2023 AA. Nazari, SA. Hosseini, and B. Taheri, "Improving the performance of differential relays in distinguishing between high second harmonic faults and inrush current," Electric Power Systems Research.

2023 B. Taheri, and A. Shahhoseini, "Direct current (DC) microgrid control in the presence of electrical vehicle/photovoltaic (EV/PV) systems and hybrid energy storage systems: A Case study of grounding and protection issue," IET Generation, Transmission & Distribution.

2023 H. Abniki, M. Hajati Samsi, B. Taheri, and SA. Hosseini, "High Impedance Fault Detection in Power Transmission Lines Using Hilbert Transform and Instantaneous Frequency," Industrial Electronic Control and Optimization.

2023 B. Taheri, SA. Hosseini and S. Salehimehr, "An overview of power swing detection method in distance relays and the factors involved," IET Generation, Transmission & Distribution.

2022 H. Abniki, M. Hajati Samsi, B. Taheri, and F. Razavi, "Power swing detection in parallel transmission lines connected to wind farms employing del2SG method," Journal of Electrical Engineering & Technology.

2022 S. Salehimehr, B. Taheri, and M. Sedighizadeh, "Short-term load forecasting in smart grids using artificial intelligence methods: A survey," The Journal of Engineering.

2022 SA. Hosseini, AA. Nazari, B. Taheri, F. Razavi, and H. Hashemi-Dezaki, "Proposing a New Approach to Generate the Differential Trajectory of the Differential Relays Using COMTRADE Files,"Sustainability.

2022 B. Taheri, SA. Hosseini, M. Sedighizadeh, and M. Khatibi, "A moving window numerical distance protection based on Flat-Top signal windowing,"Arabian Journal for Science and Engineering.

2022 B. Taheri, S. Salehimehr, and M. Sedighizadeh, "A novel strategy for fault location in shunt-compensated double circuit transmission lines equipped by wind farms based on long short-term memory,"Cleaner Engineering and Technology.

2021 B. Taheri, S. Salehimehr, and M. Sedighizadeh, "A fault-location algorithm for parallel line based on the long short-term memory model using the distributed parameter line model,"International Transactions on Electrical Energy Systems.

2021 SA. Hosseini, B. Taheri, HA. Abyaneh, and F. Razavi, "Comprehensive power swing detection by current signal modeling and prediction using the GMDH method,"Protection and Control of Modern Power Systems.

2021 B. Taheri, and M. Sedighizadeh, "A moving window average method for internal fault detection of power transformers,"Cleaner Engineering and Technology.

2021 S. Salehimehr, B. Taheri, and M. Faghihlou, "Detection of power swing and blocking the distance relay using the variance calculation of the current sampled data,"Electrical Engineering.

2020 B. Taheri, M. Faghihlou, S. Salehimehr, and F. Razavi, "A Fast Fourier Transform-Based Method for Power Swing Detection and Distance Relay Malfunction Prevention,"Journal of Control, Automation and Electrical Systems.

2020 B. Taheri, M. Faghihlou, S. Salehimehr, and F. Razavi, "Symmetrical Fault Detection During Power Swing Using Mean Value of Sampled Data from the Current Signal,"IETE Journal of Research.

2020 B. Taheri, and M. Sedighizadeh, "Detection of power swing and prevention of mal-operation of distance relay using compressed sensing theory," IET Generation, Transmission & Distribution.

2020 B. Taheri, SA. Hosseini, HA. Abyaneh, and F. Razavi, "Power swing detection and blocking of the third zone of distance relays by the combined use of empirical-mode decomposition and Hilbert transform," IET Generation, Transmission & Distribution.

2020 B. Taheri, S. Salehimehr, F. Razavi, and M. Parpaei, "Detection of power swing and fault occurring simultaneously with power swing using instantaneous frequency," Energy Systems.

2020 S. Salehimehr, B. Taheri, F. Razavi, M. Parpaei, and Milad Faghihlou "A New Power Swing Detection Method Based on Chaos Theory," Electrical Enginnering.

2019 S. Salehimehr, B. Taheri, S.A.Hosseini, H.A.Abyane, and F. Razavi,"A New Power Swing Detection Method Based on Hilbert Transform," International Journal of Industrial Electronics, Control and Optimization (IECO).

2018 B. Taheri, and F. Razavi, "Power Swing Detection Using rms Current Measurements," Journal of Electrical Engineering & Technology.

Journal papers (Persian)

2024 B. Taheri, M. Sedighizadeh, MR. Nasiri, and A. Sheikhi Finin, "Proposing a Novel Approach Non-Intrusive Load Monitoring Based on Feature Extraction Matrix and KNN Machine Learning Model,"Technovations of Electrical Engineering in Green Energy System.

2022 SA. Hosseini, and B. Taheri, "A novel strategy based on group method of data handling neural network for detection of inrush current and preventing the mal-operation of the differential relay,"Iranian Electric Industry Journal of Quality and Productivity.

2020 B. Taheri, SA. Hosseini, HA. Abyaneh, and F. Razavi, "A New Inrush Current Detection Method Based on Current Lissajous Figure,"Journal of Intelligent Procedures in Electrical Technology (JIPET).

2019 B. Taheri, SA. Hosseini, and F. Razavi, "Detection of power oscillation and simultaneous faults using Clark transform,"Iranian Electric Industry Journal of Quality and Productivity.

2019 B. Taheri, SA. Hosseini, HA. Abyaneh, and F. Razavi, "Detection of Power Swing and Blocking the Third Zone of Distance Relay Using the Norm Calculation of the Current Sampled Data,"Journal of Iranian Association of Electrical and Electronics Engineers.

Conference papers

2024 — AA. Ahmadi Pishkohi, SA. Hosseini, and B. Taheri, "A New Method Based on Empirical Wavelet Transform in Order to Detect Current Transformer Saturation in Distance Relay," 32nd International Conference on Electrical Engineering

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2020 — B. Taheri, and S. A. Hosseini, "Detection of High Impedance Fault in DC Microgrid Using Impedance Prediction Technique," 15th International Conference on Protection & Automation in Power System.

2020 — M. Faghihlou, B. Taheri, and S. Salehimehr, "High Impedance Fault Detection in DC Microgrid using Real and Imaginary Components of Power,"2020 28th Iranian Conference on Electrical Engineering (ICEE).

2020 — B. Taheri, S. A. Hosseini, and S. Salehimehr, "An Energy Variation-Based Method for Discrimination Between the Internal Fault and Inrush Current in Power Transformers," 14th International Conference on Protection & Automation in Power System.

2020 — B. Taheri, S. A. Hosseini, S. Salehimehr, and F. Razavi, "A New Method for the End-To-End Testing of Differential Relays," 14th International Conference on Protection & Automation in Power System.

2019 — B. Taheri, S. A. Hosseini, S. Salehimehr, and F. Razavi, "A Novel Approach for Detection High Impedance Fault in DC Microgrid," 34th International Power System Conference.

2019 — B. Taheri, A. Bazhdar, S. Salehimehr, M. Faghihlou, and F. Razavi, "A New Inrush Current Detection Method for Transformer Differential Protection Using Instantaneous Frequency," 12th International Conference on Engineering and Technology.

2019 — S. Salehimehr, B. Taheri, F. Razavi, and M. Faghihlou, "Power Swing Detection Using the Rate of Variations in Phase Angle of Current Signal," 12th International Conference on Engineering and Technology.

2019 — B. Taheri, S. Salehimehr, and F. Razavi, "A New Method for Fast Power Swing Detection Using the Rate of Change of Energy in the Current Signal," The 27th Iranian Conference on Electrical Engineering.

2019 — S. Salehimehr, F. Razavi, B. Taheri, and M. Parpaei, "Power Swing Detection Using Rate of Change in Negative Sequence Component of Apparent Power," 2019 International Conference on Protection and Automation of Power System (IPAPS).

2019 — B. Taheri, S. A. Hosseini, H. Askarian-Abyaneh, and F. Razavi, "A New Method for Remote Testing Distance Relay Using Internet Protocol Version 4," 2019 International Conference on Protection and Automation of Power System (IPAPS).

2019 — B. Taheri, F. Razavi, and S. Salehimehr, "Power Swing Detection Using the Variation Rates of the Average Value of Apparent Power," 2019 International Conference on Protection and Automation of Power System (IPAPS).

2018 — M. Parpaei, B. Taheri, M. M. Ghalesefidi, F. Razavi, and S. Salehimehr, "A Novel Strategy for Maximum Power Point Tracking of Fuel Cell Based on Fuzzy Controller," The 4th National Conference on Applied Research in Electrical, Mechanical, Computer and IT Engineering.

Honers

2024	Elevated to Senior Member, IEEE
2024	First Place Golpayegan Idea Show 2024
2024	3rd Place Robocup IranOpen 2024, Junior Demo-Idea League (Leader of students).
2022	The first researcher among all university students in the Qazvin province.
2020	The first researcher among all university students in the Qazvin province.
2018	Member of young researchers and elite club.

Project

2024	Conducting studies for the optimal placement of small-scale generators in the Gilan Regional Electric Company and developing the relevant software within a program created using Python and DIgSILENT.
2019	Developing a prototype laboratory for the addressable fire alarm system and creating its industrial prototype.
2017	Testing the protection relays in Iranshahr power plant.
2016	Testing the protection relays in North Yaran oil field development.
2016	Testing the protection relays in Taleqan hydroelectric dam.
2015	Simulation of transient state for Vebko Amirkabir company.

Reviewer Being

Journal	IET Generation, Transmission & Distribution	Reviewer
Journal	IET Science, Measurement & Technology	Reviewer

Journal	Electric Power Systems Research	Reviewer
Journal	Measurement	Reviewer
Journal	Energy Systems	Reviewer
Journal	International Transactions on Electrical Energy Systems	Reviewer
Journal	Energy Science & Engineering	Reviewer
Journal	Electronics Letters	Reviewer
Journal	IEEE Sensors Journal	Reviewer
Journal	IEEE Access	Reviewer
Journal	Journal of Electrical Engineering & Technology	Reviewer
Journal	Electric Power Components and Systems	Reviewer
Journal	Cogent Engineering	Reviewer
Journal	International Journal of Emerging Electric Power Systems	Reviewer
Journal	Journal of Electrical Systems and Information Technology	Reviewer
Conference	International Symposium on Hydrogen Energy, Renewable Energy and Materials (2019)	Reviewer

Licenses & certifications

2023 cs50X — Harvard University
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2023 Machin Learning with Python — Maktabkhooneh
MK-HAIMOY

References

Dr. Mostafa Sedighizadeh — Shahid Beheshti University
Associate Professor (m_sedighi@sbu.ac.ir)

Dr. Farzad Razavi — Qazvin Islamic Azad University(QIAU)
Assistant Professor(farzad.razavi@qiau.ac.ir)

Language

Persian Native

English Intermediate